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Presentation on SMART AI AUTOFILLER

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➤ Project Overview

- **Smart AI Autofill System** is an AI-powered form automation application.
- It helps users **fill forms automatically** on websites and Android applications
- Stores personal information in **secure, customizable profiles**.
- Uses **Accessibility Service and AI Field Recognition** to detect form fields.
- Supports **OCR document scanning** for Aadhaar, PAN, Marksheet, Passport, etc.
- Automatically extracts and saves information from scanned documents.
- Provides a **floating Autofill Bubble** for one-tap form filling.
- Uses **on-device AI and ML Kit** for accurate field matching and multilingual support.
- Ensures **privacy and security** through encrypted local storage, PIN lock, and biometric authentication.
- Reduces manual typing, saves time, and minimizes data entry errors.

➤ Problem Statement

- Users repeatedly enter the same personal information in multiple online forms.
- Manual form filling is time-consuming and prone to errors.
- Different websites and applications require similar data repeatedly.
- Existing autofill solutions have limited support across all apps and forms.
- Uploading information from documents such as Aadhaar, PAN, and marksheets is often a manual process.
- Managing multiple personal, academic, and professional records can be difficult.
- Data privacy and security concerns exist when using cloud-based autofill services.
- There is a need for a secure, intelligent, and universal form-filling solution.
- A smart, AI-powered autofill system is required to automate form completion efficiently.

➤ Existing System & Limitations

- Traditional form filling requires users to manually enter the same information repeatedly.
- Most existing autofill solutions work only on supported websites and applications.
- Users must switch between documents and forms to copy personal information.
- Manual data entry increases the chances of typing mistakes and incomplete submissions.
- Existing systems provide limited support for document-based data extraction.
- Many autofill tools depend on cloud services, raising privacy concerns. Managing academic, personal, and professional data across multiple platforms is difficult.
- Most applications do not offer intelligent field recognition for all types of forms.
- Current solutions lack a unified, secure, and AI-powered form automation platform.

➤ Proposed Solution

- AI-powered form automation platform for websites and Android applications.
- Automatically detects and identifies form fields using AI and Accessibility Service.
- Stores user information in secure, customizable profiles.
- One-tap autofill through a floating Autofill Bubble.
- OCR-based document scanning for Aadhaar, PAN, Passport, and Marksheets.
- Automatically extracts and saves information from scanned documents.
- Supports multilingual form recognition and field matching. Ensures privacy through encrypted local storage, PIN lock, and biometric security.
- Reduces manual data entry time, improves accuracy, and enhances user productivity.

➤ Objectives

- To automate form filling across websites and Android applications.
- To reduce the time required for repetitive data entry. To accurately detect and identify form fields using AI.
- To provide one-tap autofill through a floating Autofill Bubble.
- To extract user information from documents using OCR technology.
- To securely store personal data using encryption, PIN, and biometric authentication.
- To support multilingual form recognition and data matching.
- To minimize manual errors during form submission.
- To provide a privacy-focused and user-friendly autofill solution.

➤ System Architecture

- User interacts with the Smart AI Autofill application.
- User profiles and personal data are stored in a secure local database.
- OCR Scanner extracts information from documents such as Aadhaar, PAN, Passport, and Marksheets.
- Accessibility Service detects form fields from websites and Android applications.
- AI Field Recognition identifies the type of each form field.
- Autofill Manager retrieves matching data from the selected profile.
- Form fields are automatically filled using stored user information.
- Encryption, PIN Lock, and Biometric Authentication protect sensitive data.
- System supports multilingual form recognition and offline profile access.

➤ Technologies Used

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- **Kotlin** for Android application development.
- **Jetpack Compose** for modern and responsive user interface design.
- **Room Database (SQLite)** for local data storage and profile management.
- **Android Accessibility Service** for detecting and autofilling form fields.
- **Google ML Kit OCR** for extracting data from documents.
- **ML Kit Language Identification** for multilingual form detection.
- **ML Kit Translation API** for offline language translation support.
- **EncryptedSharedPreferences** for secure data and PIN storage.
- **CameraX** for document scanning and image capture.
- **Git and GitHub** for version control and project management.

➤ Working Methodology

- *User launches the Smart AI Autofill application.*
- *User creates or selects a profile containing personal information.*
- *User can scan documents using OCR to extract data automatically.*
- *User opens a website or application containing a form.*
- *User taps the floating Autofill Bubble.*
- *Accessibility Service detects visible form fields.*
- *AI Field Recognition identifies field labels and types.*
- *Autofill Manager retrieves matching data from the selected profile.*
- *Form fields are filled automatically and securely.*
- *User reviews the information and submits the form successfully.*

Results & Key Findings

- Form filling time is significantly reduced through automation.
- One-tap autofill provides faster data entry than manual methods.
- AI-based field recognition improves autofill accuracy.
- OCR scanning successfully extracts data from documents such as Aadhaar, PAN, and Marksheets.
- Encrypted local storage enhances data privacy and security.
- Floating Autofill Bubble improves accessibility and user convenience.
- Multilingual support enables form filling across different languages.
- Automated form completion reduces human errors and repetitive work.
- The system is effective for educational, professional, and government form applications.

➤ Conclusion

- Smart AI Autofill System provides an intelligent solution for automated form filling.
- It reduces repetitive manual data entry and saves significant time.
- AI-based field recognition improves the accuracy of form completion.
- OCR technology enables automatic extraction of data from documents.
- Secure local storage, PIN lock, and biometric authentication protect user information.
- The floating Autofill Bubble offers quick and convenient form filling across applications.
- Multilingual support increases accessibility for diverse users.
- The system minimizes human errors and enhances productivity.
- It demonstrates the practical application of AI, OCR, and Accessibility Services in real-world automation.

➤ Future Scope

- Integration of advanced AI models for smarter and more accurate field recognition.
- Support for automatic OTP detection and autofill.
- Enhanced learning system to remember frequently used form-field mappings.
- Cloud backup and synchronization of profiles across multiple devices.
- Improved OCR support for regional languages and handwritten documents.
- Support for automatic dropdown, spinner, and checkbox selection.
- WebView and browser-specific autofill enhancements.
- QR code–based profile sharing and data transfer.
- Expansion into a complete AI-powered personal information management platform.



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Thank You

Queries ?